



Listen to the (climate) science! Of Boundaries and Tipping Points

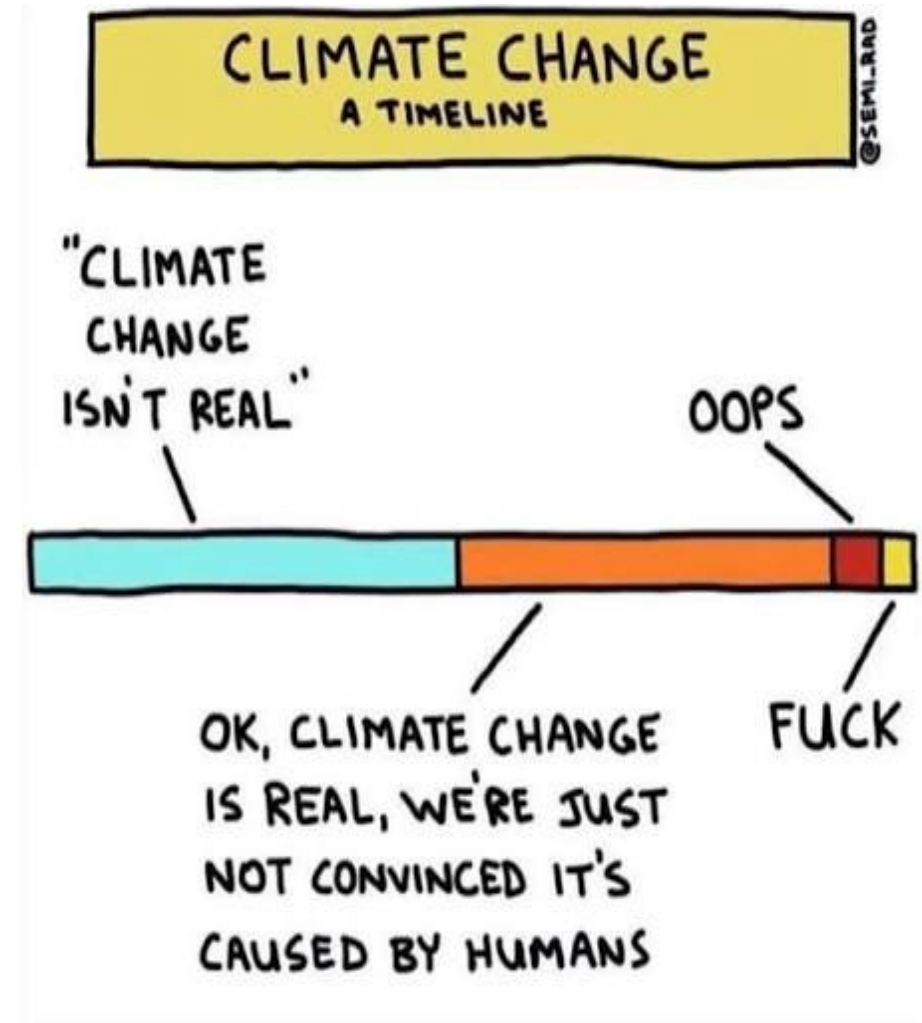
Living in the Anthropocene

14 September 2026

Laura Horn

Plan for today

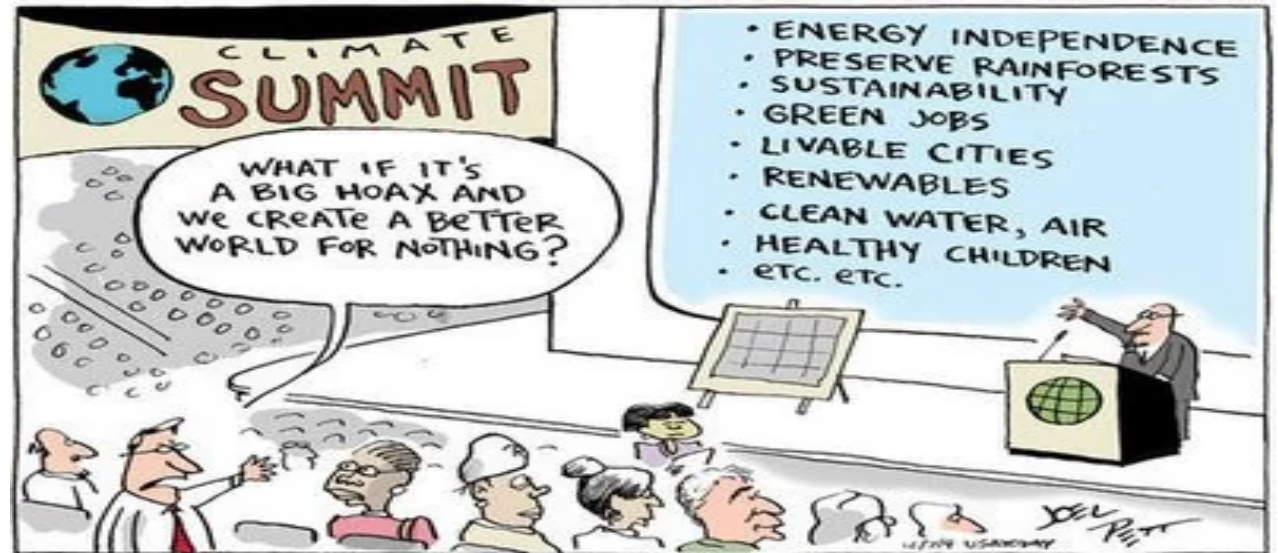
- A tour de force of climate science
 - Planetary boundaries
 - Tipping points
 - In the news right now - El Niño
- The IPCC and the role of scientists



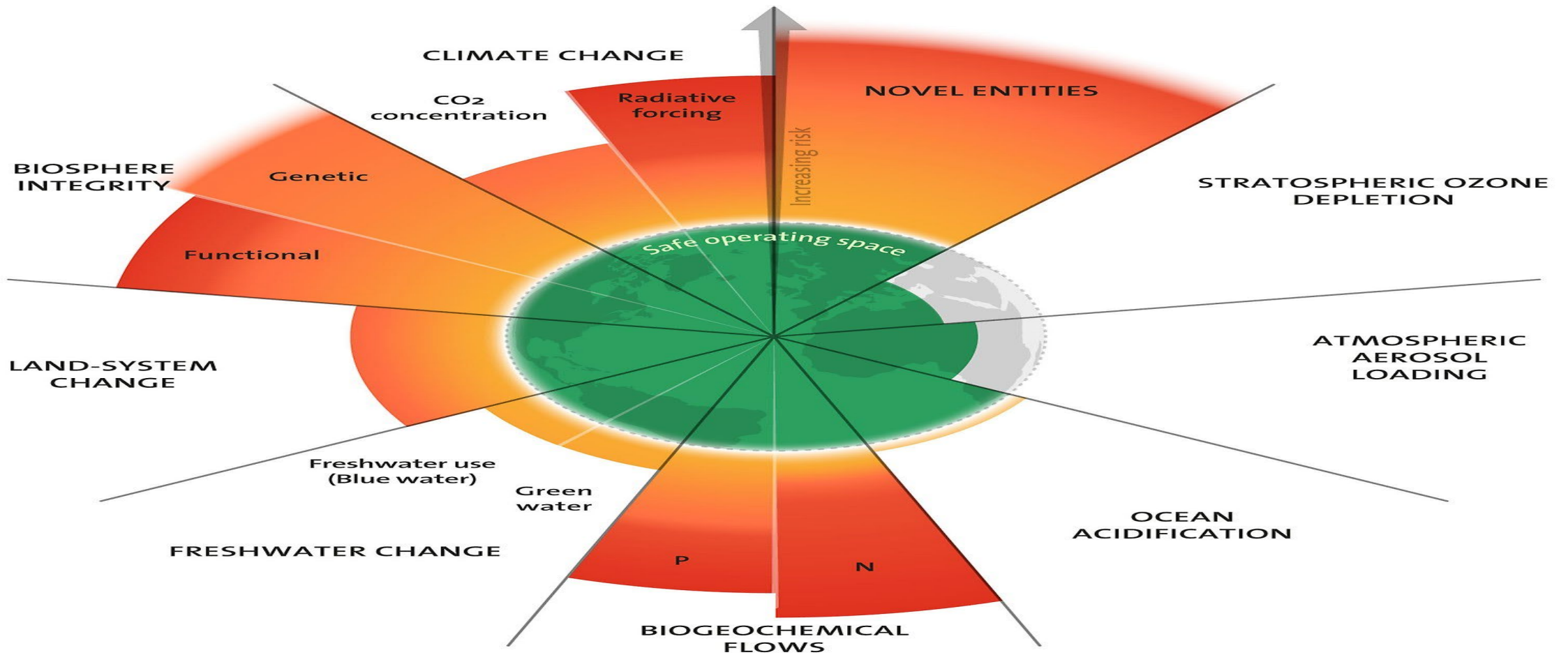
Not so fast... is this really what we should be discussing?

How to decide what the baseline should be for climate science knowledge?

Which concepts, aspects, dimensions, frameworks... should be part of 'climate literacy'?



The Planetary Boundaries framework (2025)



NB: The Biosphere boundaries will be discussed in detail in the next session!

Boundary issues – talk with your neighbour

- Turn to the person next to you.
- Pick one boundary:
 - Either the one that is close to your life,
 - Or the one you find most puzzling
- Talk to each other (5mins) – which one, and why?



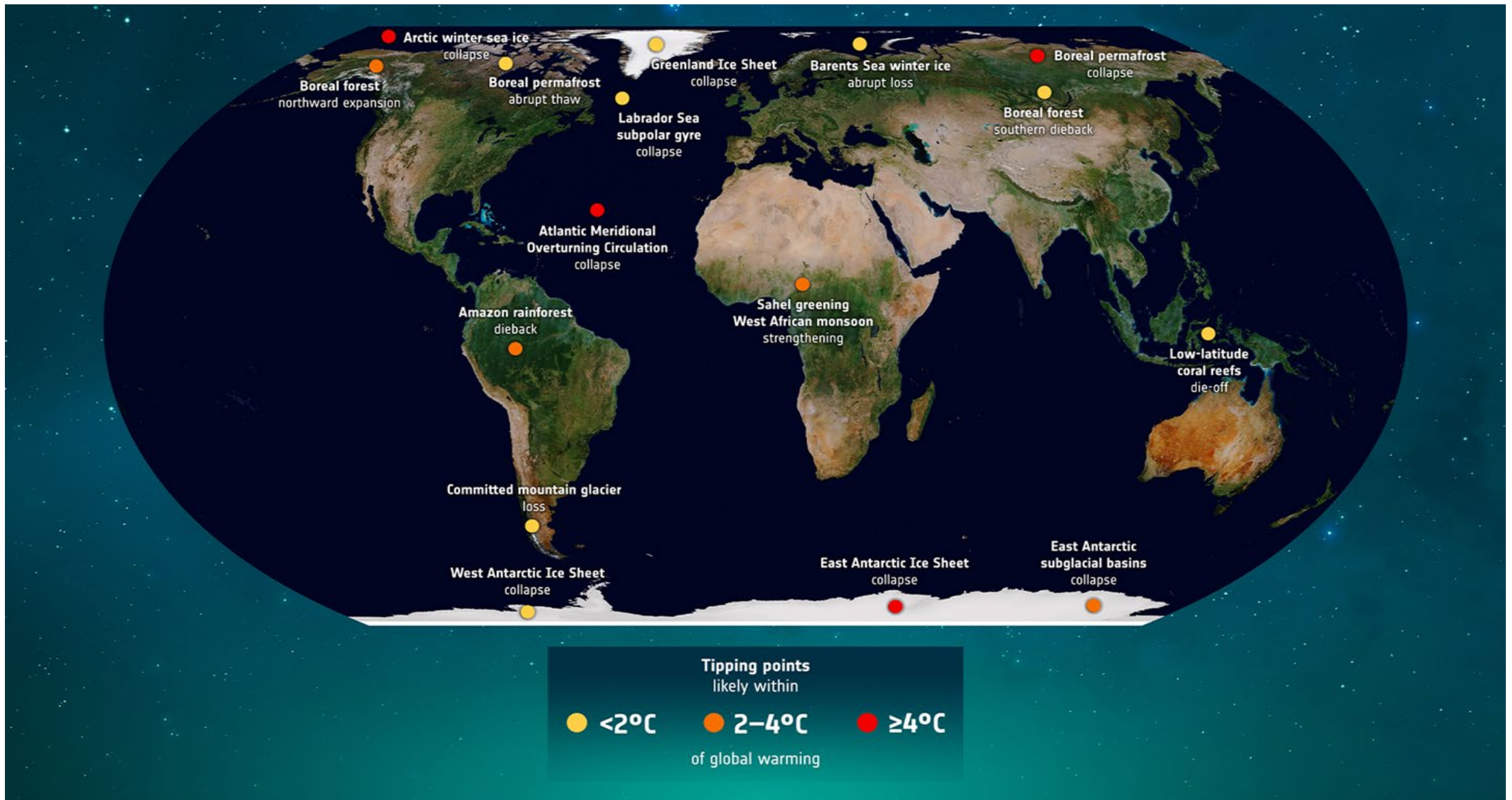
Climate Tipping points

What is a tipping point?

- a system changes into something different
- and once it starts, it keeps itself going
- Reversal is not possible

Lenton (2025): a situation within some system where amplifying feedback within that system gets so strong that it can overwhelm whatever damping feedback you have in the system, and it can get strong enough to create a self-sustaining or self-propelling change within the system.





Source: ESA (2023); see also Lenton et al (2019)

For AMOC, check out this great website! <https://amocscenarios.org>

Tipping our own points of view (ca 20mins)

We'll do a group exercise across course participants.

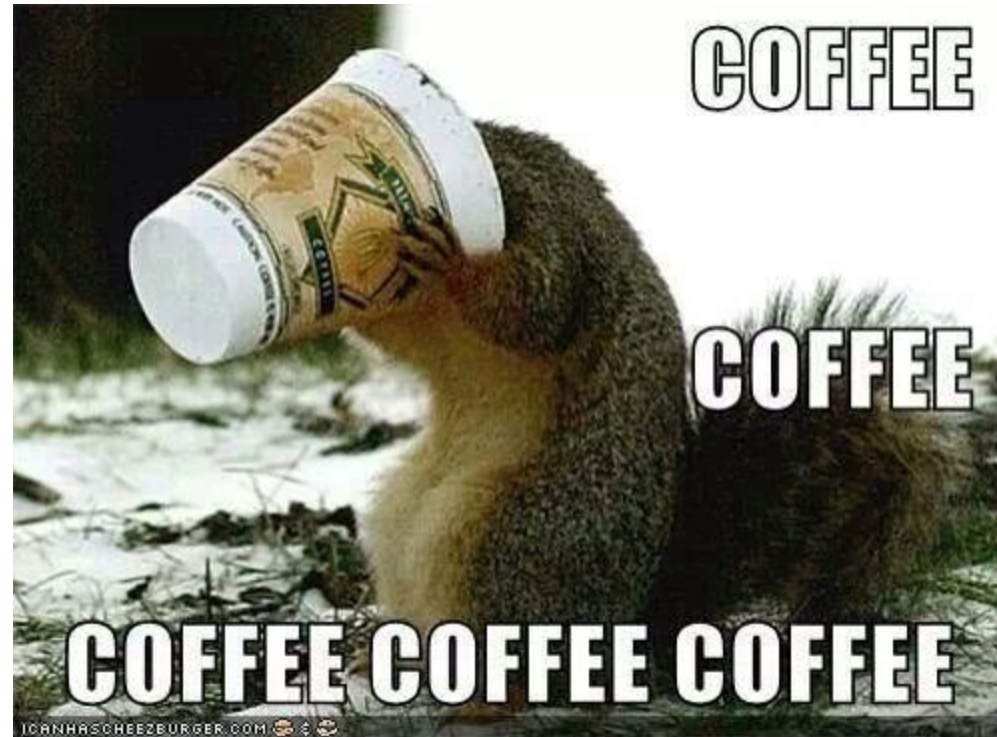
Each group will agree on **ONE tipping point** to be discussed.

Find out (*5-8mins*): what is it, what would make it tip, and how sure scientists actually are...

Then: Walk/sit and talk with your group (*fluid break!*). Discuss:

- **What would your own studies/discipline ask about this?**
- Which question(s) matter most? Which ones can't you answer?

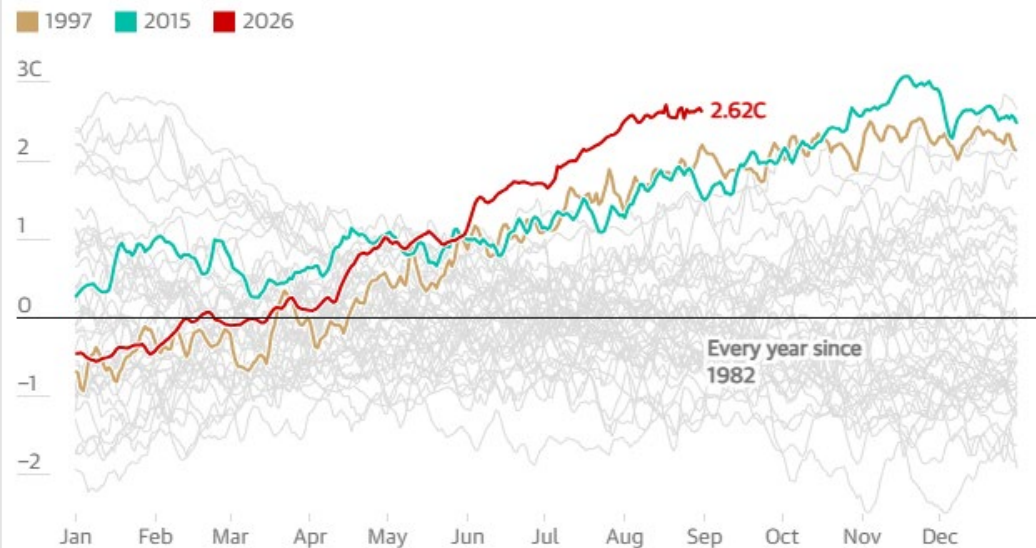
Break



El Niño – not a tipping point, still supersized

Temperature anomalies in the Niño 3.4 region of the Pacific Ocean

Daily anomaly relative to a 30-year rolling base period



Guardian graphic. Source: ERA, Climate Dashboard, Zeke Hausfather

El Niño–Southern Oscillation (ENSO)

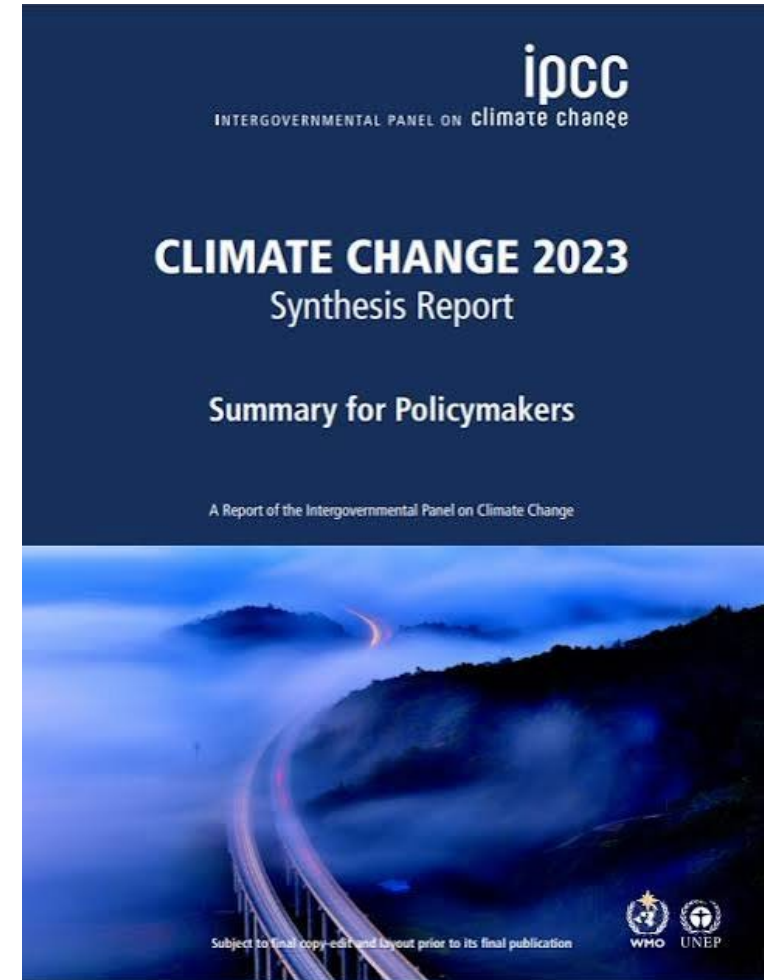
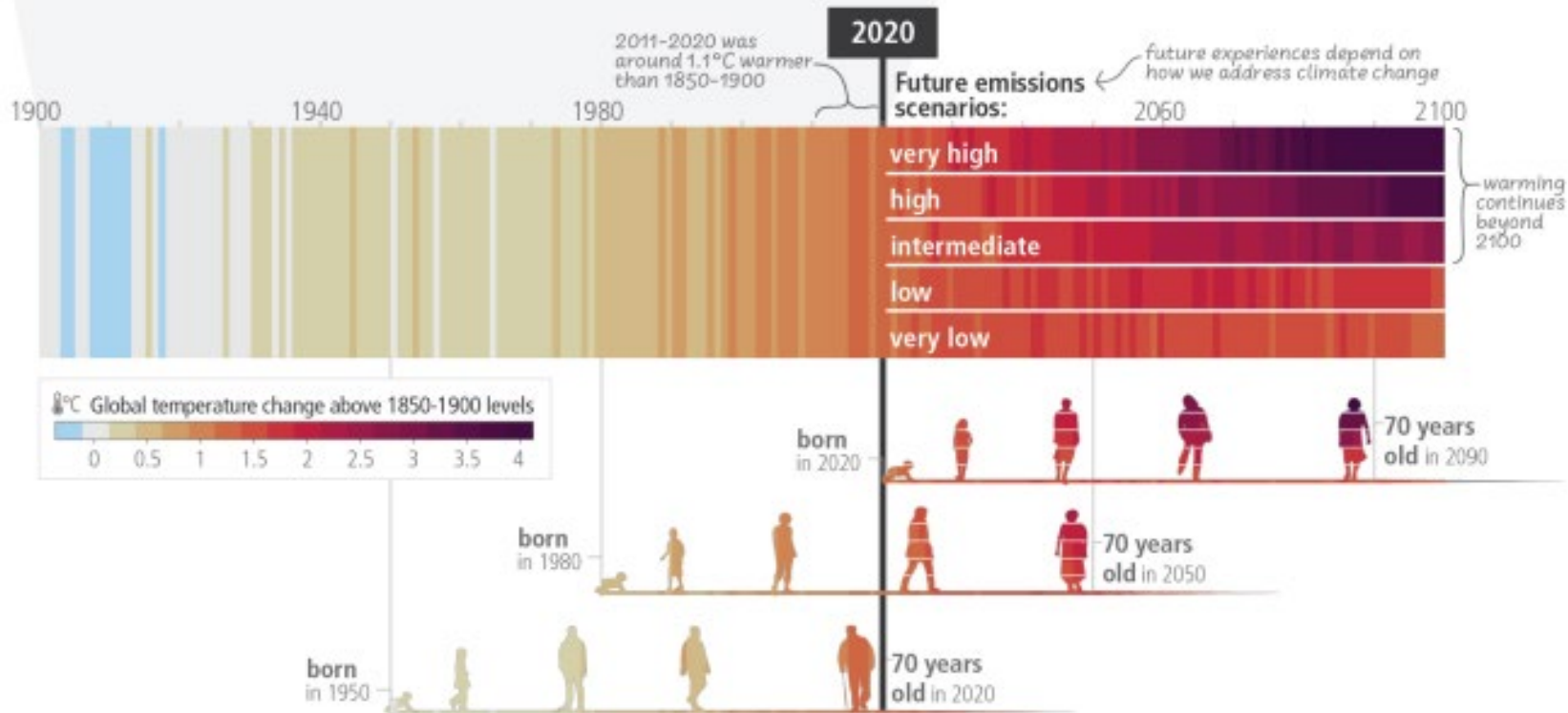
How is the El Niño phenomenon discussed?

Prediction, anticipation, uncertainty

‘Supercharged by climate change’
(2026) – attribution, responsibility, context

The IPCC

c) The extent to which current and future generations will experience a hotter and different world depends on choices now and in the near-term



— Eighth Assessment Report

ALL WORKING GROUPS

The Physical Science Basis

ipcc

INTERGOVERNMENTAL PANEL ON climate change



Adapting to Hot House Earth, the case for time-travel - IPCC

#oops

BackToTheFuture?



Let's discuss what to bring from this session

- Focus
- Agency
- Ways of Knowing

... and how to go from here?

Me when the climate anxiety kicks in

